

Client Case Study: NovaHealth Systems

Healthcare & Life Sciences | Minneapolis, MN

The Challenge

NovaHealth Systems (client ID CL-1018) operates a six-hospital regional health network across Minnesota and western Wisconsin. Following a 2020 acquisition of two community hospitals, NovaHealth found itself running four disconnected electronic health record (EHR) instances, resulting in fragmented patient records, duplicate testing, and a reported 22-minute average delay in retrieving complete patient history during emergency department intake. NovaHealth's CIO, Dr. Felix Amato, engaged CRCG in August 2022 to lead a clinical data interoperability program.

Our Approach

Phase 1: Interoperability Architecture (Months 1–3)

CRCG's Digital & Analytics team, led by Senior Engagement Partner Samuel Kestrel, designed an HL7 FHIR-based integration architecture to unify the four EHR instances behind a single clinical data repository, avoiding a disruptive full EHR consolidation in favor of a federated interoperability layer.

Phase 2: Build and Data Quality Remediation (Months 4–13)

The build phase proved more complex than initially scoped: legacy data extracts from the two acquired community hospitals contained significant data quality issues (duplicate patient identifiers, inconsistent allergy coding), requiring an unplanned data quality remediation workstream that extended the original 11-month build timeline by roughly two months.

Phase 3: Cutover and Hypercare (Months 14–17)

CRCG supported a phased go-live across all six hospitals between August and November 2023, culminating in program close-out in December 2023 with a 90-day hypercare period.

Results

Metric	Before	After	Notes
ED intake history retrieval time	22 min (avg)	4 min (avg)	Measured across all 6 hospitals post-cutover
Duplicate diagnostic test rate	11.4%	5.2%	Driven by unified record visibility
EHR instances	4 (disconnected)	4 (federated via interoperability layer)	Full EHR consolidation explicitly out of scope
Program cost (PRJ-2245)	\$2.65M (budget)	\$2.81M (actual)	Variance driven by data quality remediation workstream

“We went into this thinking we needed to rip out and replace four EHRs. CRCG showed us we didn’t — we needed better plumbing between them. That reframing alone saved us an estimated \$9 million in avoided EHR consolidation costs.”

— Dr. Felix Amato, Chief Information Officer, NovaHealth Systems